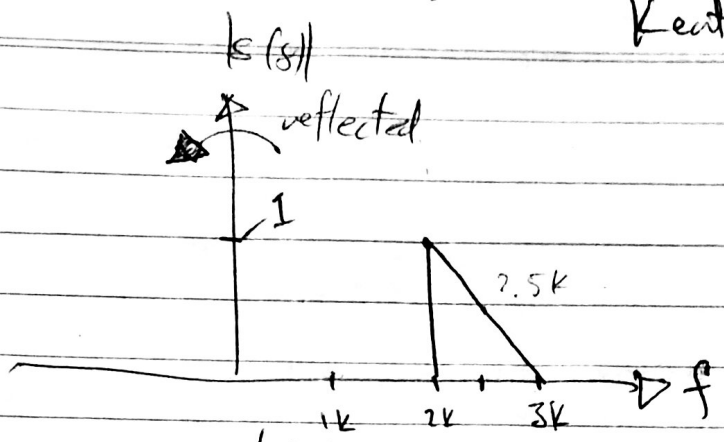
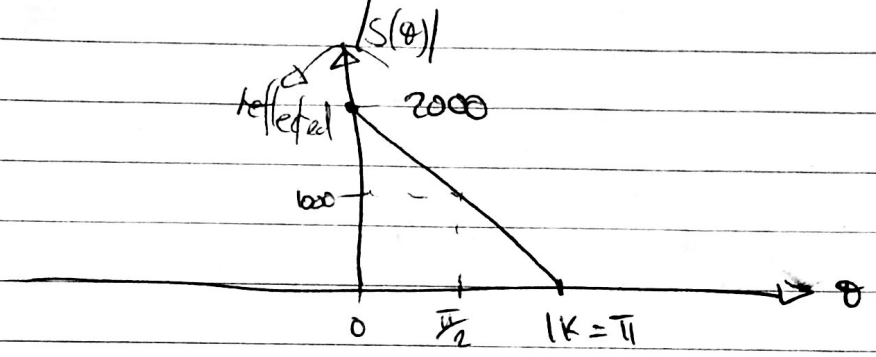
DSP ~~Post~~ exam 2015 Angus Keatinge 7/6/16

Q1 a)





b) $x(t) = 5\cos(600\pi t) + 4\cos(1800\pi t)$ 10000 bits/s

1000 samples/sec = f_s → $f_s/2 = 500$ Hz

i) 300 Hz

100 Hz

ii) $\theta = \frac{2\pi f}{f_s}$ $t = nT$ $T = \frac{1}{f_s}$

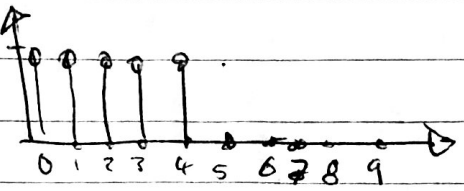
$x(n) = 5\cos\left(\frac{2\pi n}{3}\right) + 4\cos\left(\frac{9\pi}{10}\frac{9\pi n}{5}\right)$

$= 5\cos\left(\frac{2\pi n}{3}\right) + 4\cos\left(\frac{\pi n}{10}\right)$

$2\pi n\left(\frac{9}{10}\right)$

$= 2\pi n\left(\frac{9}{10}\right)$

c) i)



$$h(n) = \delta(n) + \delta(n-1) + \delta(n-2) + \delta(n-3) + \delta(n-4)$$

$$H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$$

$$= z^{2}\left((z^{-2} + z^{2}) + (z^{-1} + z^{1}) + 1\right)$$

$$H(\theta) = e^{j2\theta}\left(2\cos(2\theta) + 2\cos\theta + 1\right)$$

$$= z^{-2}\left((z^{2} + z^{-2}) + (z^{1} + z^{-1}) + 1\right)$$

$$H(\theta) = e^{-j2\theta}\left(2\cos(2\theta) + 2\cos(\theta) + 1\right)$$

OR

$$H(z) = \frac{1 - z^{-5}}{1 - z^{-1}}$$

$$H(\theta) = \frac{1 - e^{-j\theta 5}}{1 - e^{-j\theta}} = \frac{e^{-j\frac{5\theta}{2}}}{e^{-j\frac{\theta}{2}}} \times \frac{e^{j\frac{5\theta}{2}} - e^{-j\frac{5\theta}{2}}}{e^{j\frac{\theta}{2}} - e^{-j\frac{\theta}{2}}}$$

$$= e^{-j\theta\left(\frac{5}{2} - \frac{1}{2}\right)} \times \frac{2j\sin\left(\frac{5\theta}{2}\right)}{2j\sin\left(\frac{\theta}{2}\right)}$$

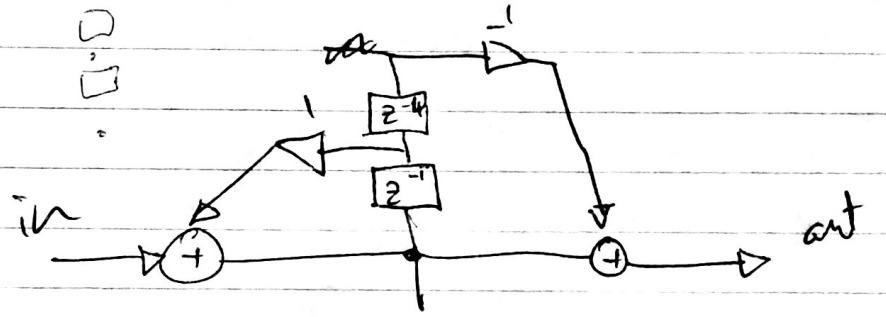
$$= e^{-j2\theta}\,\frac{\sin\left(\frac{5\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)} \quad \checkmark$$

ii)

$$H(z) = \frac{1 + z + z^2 + z^3 + z^4}{z^4}$$

$$= \frac{1 - \frac{1}{z^5}}{1 - \frac{1}{z}} = \frac{\frac{z^5 - 1}{z^5}}{\frac{z - 1}{z}} = \frac{z^5 - 1}{z^4 (z-1)}$$

$$= \frac{1 - z^{-5}}{1 - z^{-1}} = \frac{z^5 - 1}{z^5 - z^4}$$



$4 \times 4 \; 1 - 1 = 0$

1 → 1, 1, 1, 1, 1, 0, 0, 0, ...

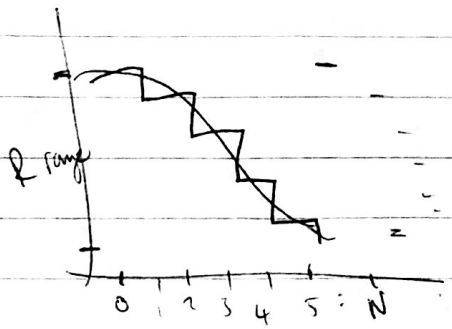
iii) $$\text{groupdelay} = \frac{d\phi(\theta)}{d\theta}$$

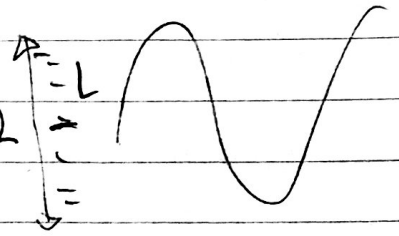
$$\phi(\theta) = -2\theta$$

$$\frac{d\phi}{d\theta} = -2 \sim \text{const.}$$

(Q2) a) i) $SQNR = 10\log(P_s) + 10.8 + 20\log\left(\frac{L}{R}\right)$

$P_s = \frac{1}{N}\sum_{n=0}^{N-1} x^2(n)$



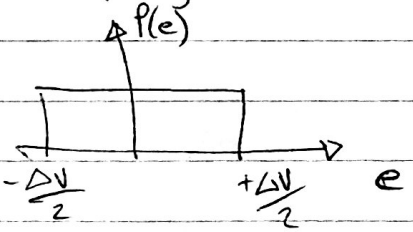


$e = \frac{\Delta V_{max}}{2}$

$\Delta V_{max} = \frac{R}{L}$

$SQNR = \frac{\text{Signal power}}{\text{Noise power}}$

Assume uniformly distributed error



$\therefore$ quantisation noise power $\sigma^2 = \frac{\Delta V^2}{12} = \frac{R^2}{12L^2}$

Signal power: assume sine, centered about 0.

$$SQNR = 10\log\left(\frac{P_s}{R^2/12L^2}\right)$$

$$= 10\log(P_s) + 10\log(12) + 10\log(L^2) - 10\log(R^2)$$

$$= 10\log(P_s) + 10.79 + 20\log(L) - 20\log(R)$$

ii) for a B bit quantiser

$L = 2^B$ $\quad x(t) = A\cos(2\pi ft)$

$R = A \rightarrow R = 2A$

$P_s = \frac{1}{N}\sum_{n=0}^{N-1} x^2(n) \leftarrow \sigma^2 = \frac{A^2}{2}$

$\log_a b = \frac{\log_{10}(b)}{\log_{10}(a)}$

$$SQNR = 20\log\left(\frac{A}{\sqrt{2}}\right) + 10.8 + 20\log(2^B) - 20\log(A)$$

$$= 10.8 + 20\log\left(\frac{1}{\sqrt{2}}\right) + 20\log(2^B)$$

$$= 4.779 + 20(\log_2(2^B) \times \log_{10}(2))$$

$$= + 6.02B \checkmark$$

$\log_2(2^B) = B = \frac{\log_{10}(2^B)}{\log_{10}(2)}$

~~$SQNR = 10\log\left(\frac{P_s \cdot 12}{(R/L)^2}\right) = 10\log\left(\frac{12 P_s L^2}{R^2}\right)$~~

$$SQNR = 20\log\left(\frac{A}{\sqrt{2}}\right) + 10.8 + 20\log(2^B) - 20\log(2A)$$

$$= 20\log\left(\frac{1}{2\sqrt{2}}\right) + 10.8 + 20\log_2(2^B) \times \log_{10}(2)$$

$$= 1.769 + 6.02B \quad (dB) \checkmark$$

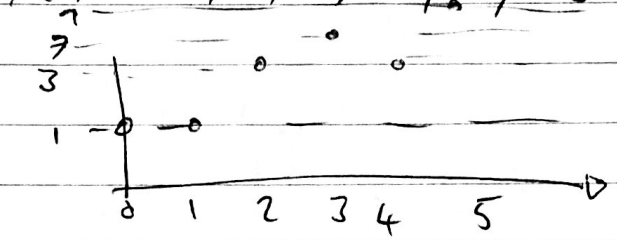
b) $h(n)$

$$y(n) = x(n) * h(n)$$

$$= [1, -1, 3] * [1, 2, 2, 3]$$

$$(3; -1, 1)$$

$$y(n) = [1, 1, 3, 7, 3, 9] \quad \checkmark$$



c) i) $H(z) = k \dfrac{1 - z^{-1}}{1 + bz^{-1}} \qquad 0 < b < 1$

$\omega = \pi \qquad e^{-j\omega} = -1$

$$H(\omega) = k \frac{1 - e^{-j\omega}}{1 + be^{-j\omega}}$$

$$= \left| k \frac{1 + 1}{1 - b} \right| = 1 \Rightarrow k = \frac{1-b}{2} \text{ or } k = \frac{b-1}{2}$$

ii) $$H(z) = \frac{1-b}{2} \frac{1 - z^{-1}}{1 + bz^{-1}}$$

$$= \frac{k}{b} \left(\frac{z^{-1}}{\frac{1}{b} + z^{-1}} - \frac{z^{-1}}{\frac{1}{b} + z^{-1}} \right)$$

$\frac{z}{z-a} \rightarrow a^n u(n)$

$z \rightarrow a = z + \frac{1}{b}$

$$h(n) = \frac{k}{b} \left(-\left(\frac{-1}{b}\right)^{-n} u(-n) + \left(\frac{-1}{b}\right)^{1-n} (u(1-n)) \right)$$

$$= \frac{2}{b - b^2} \left(\frac{1}{(-b)^{n-1}} u(1-n) - \frac{1}{(-b)^n} u(-n) \right)$$

$H(z) = k \dfrac{z-1}{z+b}$

pole at $z = -b$
inside unit circle
stable.

c) iii)

$$\theta_{3dB} \rightarrow |H(\theta_c)| = \frac{1}{\sqrt{2}}$$

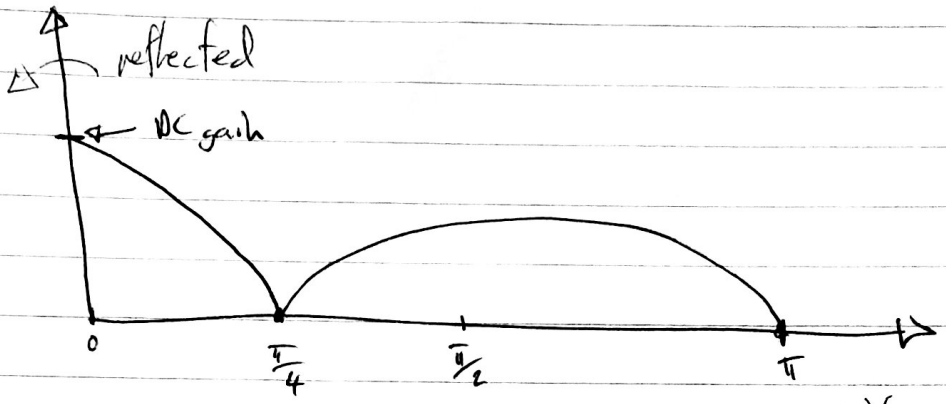
$$H(\theta) = k\,\frac{1 - e^{-j\theta}}{1 + be^{-j\theta}}$$

$$|H(\theta)|^2 = |k|^2 \times \frac{(1 - e^{-j\theta}) \times (1 + be^{j\theta})}{(1 + be^{-j\theta})(1 + be^{j\theta})}$$

$$\downarrow$$

$$|k|^2 \left| \frac{1 - e^{-j\theta_c} + be^{j\theta_c} - b}{1 + b + 2b\cos\theta_c} \right| = \frac{1}{2}$$

Q3 a) i)



ii) $H(z) = k \dfrac{(z^2+1)(z+\frac{1}{2})}{z^2(z-\frac{4}{5})}$

$(z - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}j)(z - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}j)$

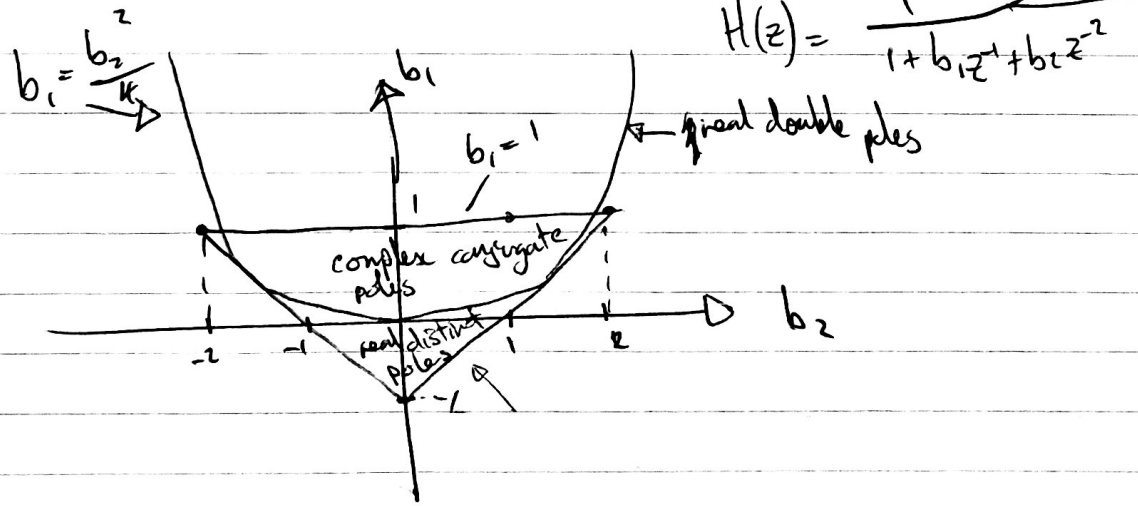
$= z^2 + \frac{1}{2} + \frac{1}{2} = (z^2+1)$

$H(\theta) = k \dfrac{(e^{j\theta}+1)(e^{j\theta}+\frac{1}{2})}{e^{2j\theta}(e^{j\theta}-\frac{4}{5})}$

$H(0) = k\left(\dfrac{2 \times \frac{3}{2}}{\frac{1}{5}}\right) = k \times 15 = 15\left(1 - \frac{1}{\sqrt{2}}\right)$

$H(z) = \left(1 - \frac{1}{\sqrt{2}}\right) \dfrac{(z^2+1)(z+\frac{1}{2})}{z^2(z-\frac{4}{5})}$

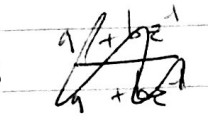
b)



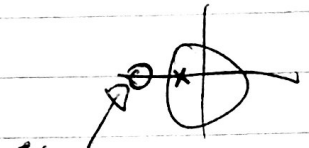
$H(z) = \dfrac{1}{1 + b_1 z^{-1} + b_2 z^{-2}}$

c) $H(z) = \dfrac{14 + 11z^{-1}}{9 + 6z^{-1}}$ ~ parallel connection of two all pass filters.

$$= \frac{9 + 6z^{-1}}{9 + 6z^{-1}} + \frac{5 + 5z^{-1}}{9 + 6z^{-1}}$$

All pass 

$$= \frac{8 + 8z^{-1}}{9 + 6z^{-1}} + \frac{12 + 9z^{-1}}{9 + 6z^{-1}}$$



$6 + 9z^{-1}$ on top of $2 + 3z^{-1}$

$$= \frac{1}{3} \times \frac{6 + 9z^{-1}}{9 + 6z^{-1}} + \frac{3}{4} \frac{9 + 6z^{-1}}{9 + 6z^{-1}}$$

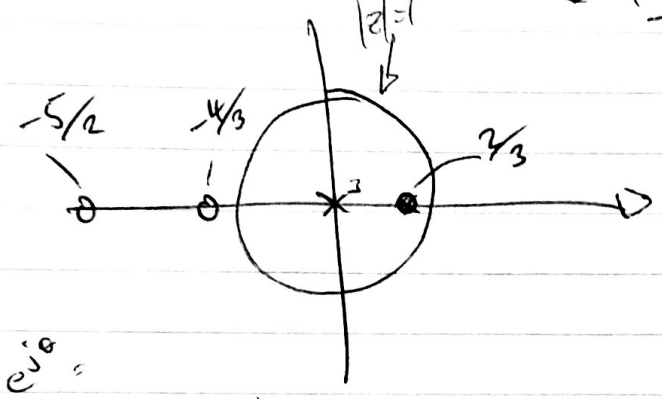
$$= \frac{1}{3} \times \frac{6 + 9z^{-1}}{9 + 6z^{-1}} + \frac{3}{4} \times 1 \quad \checkmark$$

$G(0^\circ) = (9+6-8)(2+5)$

$= 7 \times 7 = 49$

d) $G(z) = (9+6z^{-1}-8z^{-2})(2+5z^{-1})$

$= \frac{(9z^2+6z-8)(2z+5)}{z^3}$



$\frac{-6 \pm \sqrt{6^2+4 \cdot 8 \cdot 9}}{2 \cdot 9} = \frac{-6 \pm 18}{18}$

$= -\frac{1}{3} \pm 1$

$= \frac{2}{3}, -\frac{4}{3}$

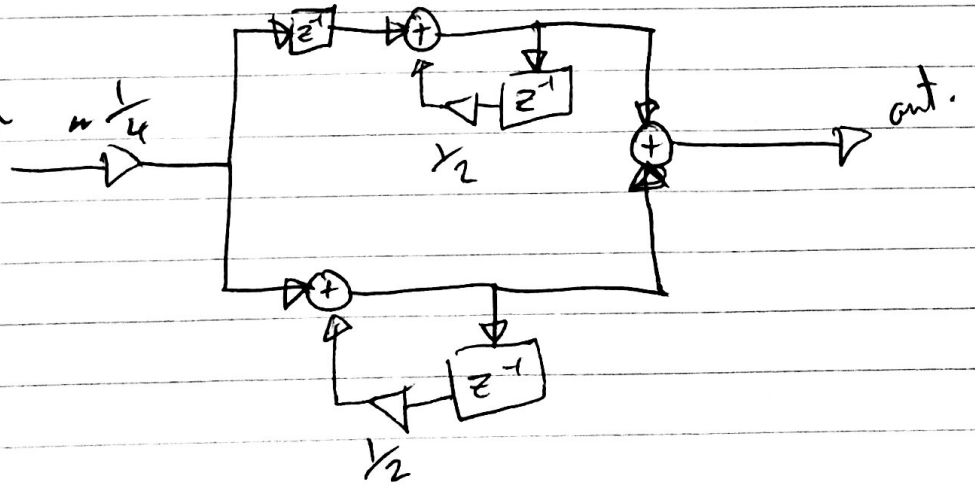
$e^{j\theta}$

- $G_{min}(z) = k_{min}\frac{(z-\frac{2}{3})(z+\frac{3}{4})(z+\frac{2}{5})}{z^3}$ $\quad k_{min} = 60$

$G_{max}(z) = k_{max}\frac{(z-\frac{3}{2})(z+\frac{4}{3})(z+\frac{5}{2})}{z^3}$ $\quad k_{max} = -12$

e) $H(z) = \frac{1}{4}\frac{1+z^{-1}}{1-0.5z^{-1}}$

$= \frac{1}{4}\left(\frac{1}{1-0.5z^{-1}} + \frac{z^{-1}}{1-0.5z^{-1}}\right)$



Q4

$$H(s) = \frac{0.1s}{s^2 + 0.1s + 1}$$

$f_s = 2400\,Hz$

$f_a = 600\,Hz$

$\theta_{res} = \pi/2$

a) i) $H(\frac{s}{z}) = \frac{2}{10T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)$

first $s \to \frac{s}{\omega_c}$

this is your θ_c

$$\omega_c = \frac{2}{T}\tan\left(\frac{\theta}{2}\right) = 2 \times 2400 \times \tan\left(\frac{\pi}{2\epsilon}\right)$$

$$= 4800 = \frac{2}{T}$$

$$H\left(\frac{s}{\omega_c}\right) = \frac{0.1\, s\left(\frac{2T}{\pi/2}\right)}{s^2\left(\frac{T}{2}\right)^2 + 0.1s\left(\frac{T}{2}\right) + 1}$$

$$H(z) = \frac{0.1\left(\frac{z}{T}\right)\left(\frac{T}{2}\right)\left(\frac{1-z^{-1}}{1+z^{-1}}\right)}{\left(\frac{T}{2}\right)^2\left(\frac{z}{T}\right)^2\left(\frac{1-z^{-1}}{1+z^{-1}}\right)^2 + 0.1\left(\frac{1-z^{-1}}{1+z^{-1}}\right)\left(\frac{T}{2}\right)\left(\frac{z}{T}\right) + 1}$$

$$= \frac{\frac{1}{10}(1-z^{-1})(1+z^{-1})}{\left(\frac{1-z^{-1}}{1+z^{-1}}\right)^2 + \frac{1}{10}(1+z^{-1})(1-z^{-1}) + (1+z^{-1})^2}$$

$$= \frac{1}{10} \times \frac{(1-z^{-1})(1+z^{-1})}{2 + 2z^{-2} + \frac{1}{10} - \frac{1}{10}z^{-2}}$$

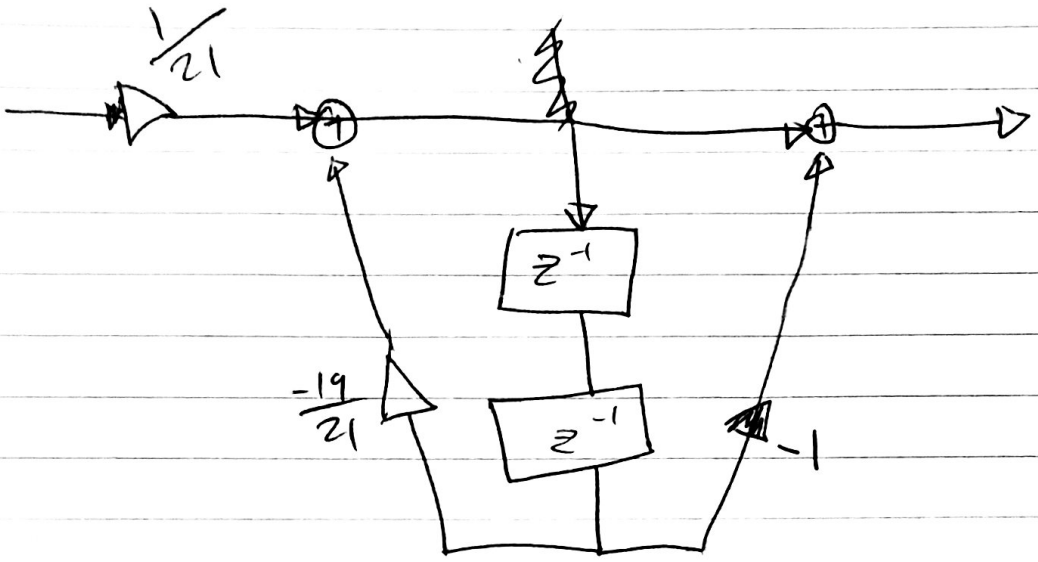
$\frac{21}{10} \times \frac{10}{21}$

$\frac{19}{10} \times \frac{10}{21} = \frac{19}{21}$

$a_0 = \frac{1}{21}$

$b_1 = \frac{19}{21}$

ii)



b) i) $y(n) = a\,y(n-1) + x(n)$ $\qquad a = e^{j\theta_0}$

$= e^{j\theta_0} y(n-1) + x(n)$

$e^{j\theta_0} = \cos\theta_0 + j\sin\theta_0$

$y(n) = y_R(n) + j\,y_i(n)$

~~$y(n) = \cos\theta_0\, y(n-1) + x(n) + j(\sin\theta_0\, y(n-1))$~~

~~$y_R(n) = \cos\theta_0\, y(n-1) + x(n)$~~

$y_i(n) = \sin($

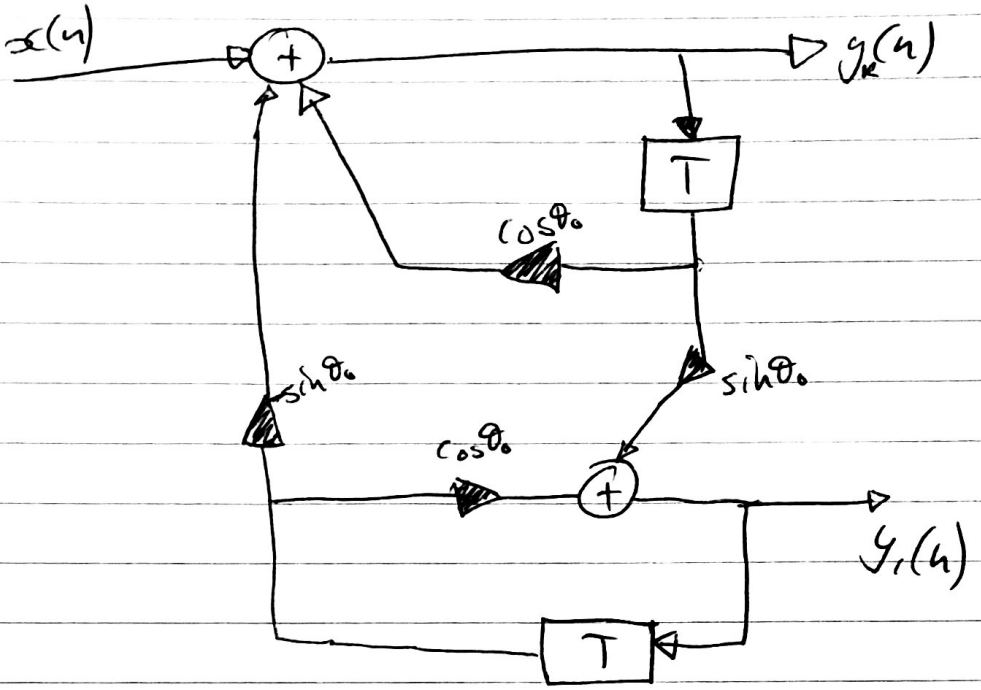
$y_R(n) + j\,y_i(n) = a\,y_R(n-1) + aj\,y_i(n-1) + x(n)$

$= \cos\theta_0\, y_R(n-1) + j\sin\theta_0\, y_R(n-1) + j\cos\theta_0\, y_i(n-1) - \sin\theta_0\, y_i(n-1) + x(n)$

$y_R(n) = x(n) + \cos\theta_0\, y_R(n-1) - \sin\theta_0\, y_i(n-1)$

$y_i(n) = \sin\theta_0\, y_R(n-1) + \cos\theta_0\, y_i(n-1)$

ii)



iii) if $x(n) = \delta(n)$ then $y_R(n) = \cos(n\theta_0)\, a(n)$

$$Y(z) = Y(z)\, a\, z^{-1} + X(z$$

$$\frac{Y(z)}{X(z)} = \frac{1}{1 - a z^{-1}} \qquad a = e^{-j\theta_0}$$

$$= \cos\theta_0 + j\sin\theta_0$$

$$H(z) = \frac{1}{1 - a z^{-1}}$$

$$= \frac{1}{1 - \cos\theta_0 e^{-j\theta_0} z^{-1}} \quad \frac{(1 - e^{j\theta_0} z^{-1})}{(1 - e^{j\theta_0} z^{-1})}$$

$$= \frac{1 - e^{j\theta_0} z^{-1}}{1 + z^{-2} - z^{-1}(e^{j\theta_0} + e^{-j\theta_0})}$$

$$e^{j\theta_0} + e^{-j\theta_0} = 2\cos\theta_0$$

$$H(z) = \frac{z^2(1 - \cos\theta_0 \cdot z^{-1} - j\sin\theta_0 z^{-1})}{z^2 - z\, 2\cos\theta_0 + 1}$$

$$H(z) = \frac{z^2 - z\cos\theta_0}{z^2 - 2\cos\theta_0 z + 1} - j\frac{\sin\theta_0 z}{z^2 - 2\cos\theta_0 z + 1}$$

$$H_R(z) = \frac{z^2 - z\cos\theta_0}{z^2 - 2\cos\theta_0 z + 1} \longrightarrow h_R(n) = \cos(\theta_0 n)\,u(n)$$

$$H_I(z) = \frac{-\sin\theta_0 z}{z^2 - 2\cos\theta_0 z + 1} \longrightarrow h_I(n) = -\sin(\theta_0 n)\,u(n)$$

as required.

not quite right but pretty fukn close

c) $y(n) = x(n) - x(n-4)$

$Y(z) = X(z) - X(z)z^{-4}$

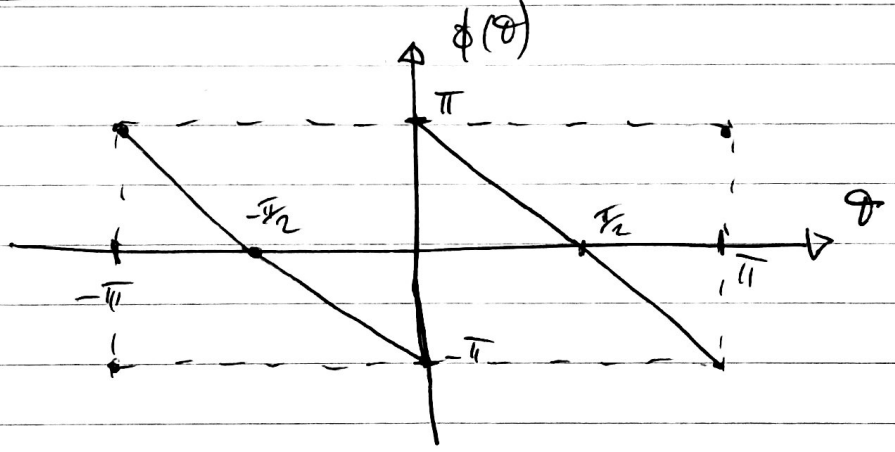
$H(z) = 1 - z^{-4}$

$H(\theta) = 1 - e^{-4j\theta}$

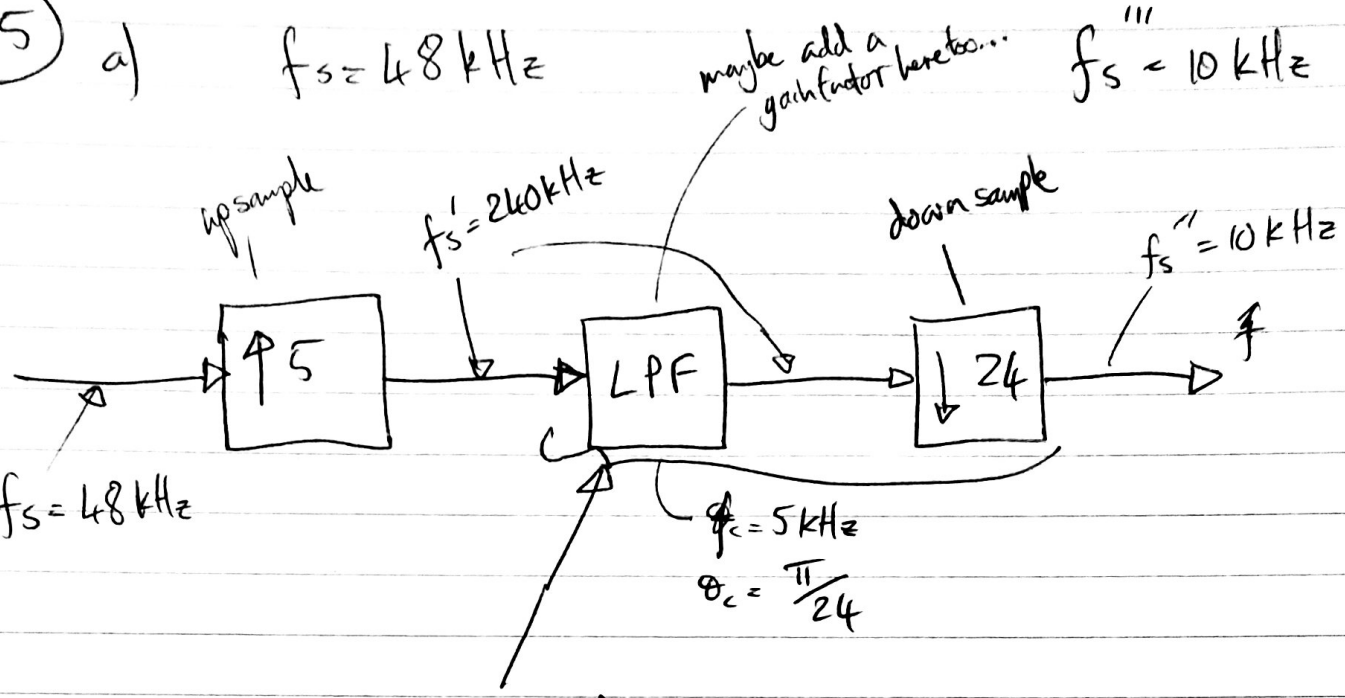
$= e^{-2j\theta}\left(e^{2j\theta} - e^{-2j\theta}\right)$

$= e^{-2j\theta}\, 2j\,(\sin 2\theta) \qquad j = e^{-j\pi}$

$\therefore \boxed{\phi(\theta) = -2\theta - \pi}$

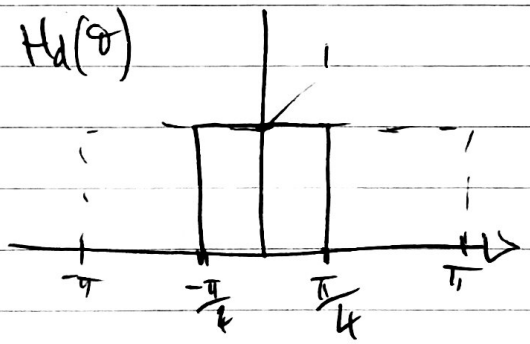


(Q5) a) $f_s = 48\,kHz$



If the LPF is an FIR filter, then these two steps can be done in one go, minimising complexity.

b) i) $x(n) = \frac{1}{2}\delta(n+2) + \delta(n) + \frac{1}{2}\delta(n-2)$



$X(z) = \frac{z^{+2}}{2} + 1 + \frac{z^{-2}}{2}$

$X(\theta) = 1 + \cos 2\theta$

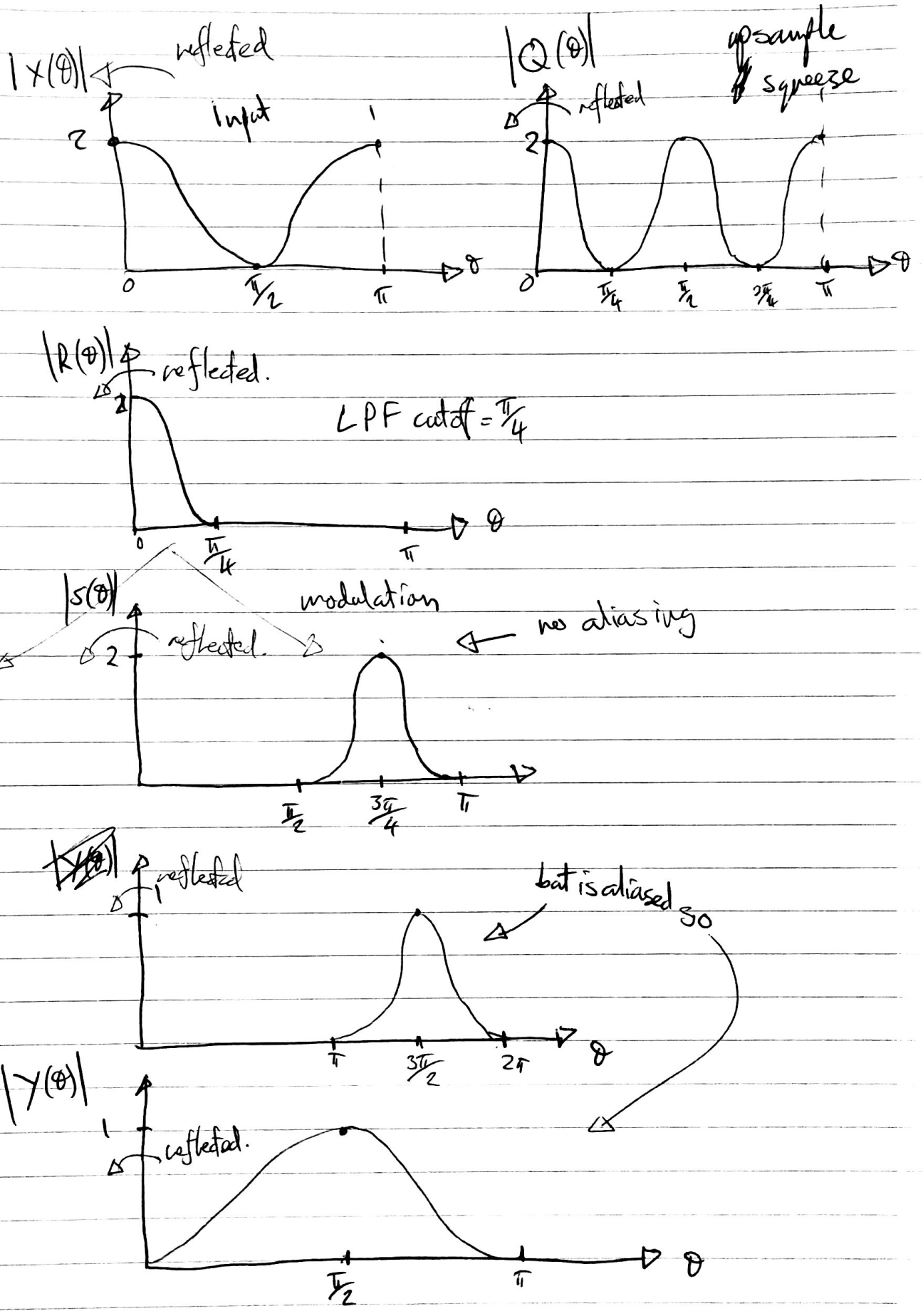
$Q(n) = \frac{1}{2}\delta(n+4) + \delta(n) + \frac{1}{2}\delta(n-4)$

inserted zeros here — inserted two zeros here

$Q(z) = 1 + \frac{1}{2}(z^4 + z^{-4})$

$Q(\theta) = 1 + \cos 4\theta$ as required

ii)


|X(θ)|
reflected
input
2
1
0
π/2
π
θ
|Q(θ)|
upsample
squeeze
reflected
2
0
π/4
π/2
3π/4
π
θ
|R(θ)|
reflected.
2
LPF cutoff = π/4
0
π/4
π
θ
|S(θ)|
modulation
no aliasing
reflected.
2
π/2
3π/4
π
|Y(θ)|
reflected
1
but is aliased so
π
3π/2
2π
θ
|Y(θ)|
1
reflected.
π/2
π
θ

c)

$X(\theta) \in$ Real $\therefore$ it has a

$X(\theta) = e^{0j} \, |X(\theta)|$

$H(z) = \frac{1}{2} + z^{-1} - \frac{1}{2} z^{-2}$

$= z^{-1}\left(\frac{z}{2} + 1 - \frac{z^{-1}}{2}\right)$

$H(\theta) = e^{-j\theta}\left(1 + j\frac{e^{j\theta} - e^{-j\theta}}{2j}\right)$

$= e^{-j\theta}(1 + j\sin\theta)$

$G(z) = \frac{1}{2} + z^{1} - \frac{1}{2} z^{2}$

$= z^{1}\left(1 + \frac{z^{-1}}{2} - \frac{z}{2}\right)$

$G(\theta) = e^{j\theta}(1 - j\sin\theta)$

$Y(\theta) = G(\theta)\, H(\theta)\, X(\theta)$ — complex conjugates.

$= X(\theta)\left[e^{j\theta} e^{-j\theta}\right]\left[(1 + j\sin\theta)(1 - j\sin\theta)\right]$

all real all real all real

$\therefore Y(\theta) =$ all real $\therefore$ has zero phase response